

120 Days of Quantitative Finance

Month 3: Machine Learning and Deep Learning

Week 12: Deep Learning

Day 78: Neural Networks and Backpropagation

1 Introduction

Neural networks are a class of machine learning models inspired by the structure of the human brain. They are widely used in modern quantitative finance for tasks such as return prediction, volatility forecasting, and pattern recognition.

A neural network consists of layers of interconnected nodes (neurons) that transform inputs into outputs through a series of nonlinear operations.

2 Structure of a Neural Network

A simple feedforward neural network consists of:

- Input layer
- Hidden layer(s)
- Output layer

Let $x \in \mathbb{R}^n$ be the input vector.

For a single hidden layer:

$$z^{(1)} = W^{(1)}x + b^{(1)}$$

$$a^{(1)} = \phi(z^{(1)})$$

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$$

$$\hat{y} = a^{(2)} = \phi(z^{(2)})$$

where:

- $W^{(1)}, W^{(2)}$ are weight matrices
- $b^{(1)}, b^{(2)}$ are bias vectors
- $\phi(\cdot)$ is an activation function

3 Loss Function

To train the network, we define a loss function that measures prediction error.

For regression problems:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The objective is to minimize this loss by adjusting the weights and biases.

4 Forward Propagation

Forward propagation computes the output of the network:

$$x \rightarrow z^{(1)} \rightarrow a^{(1)} \rightarrow z^{(2)} \rightarrow \hat{y}$$

This is simply a sequence of linear transformations followed by nonlinear activations.

5 Backpropagation

Backpropagation is the algorithm used to compute gradients of the loss function with respect to the network parameters.

Using the chain rule:

$$\frac{\partial L}{\partial W^{(2)}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z^{(2)}} \cdot \frac{\partial z^{(2)}}{\partial W^{(2)}}$$

Similarly, for the first layer:

$$\frac{\partial L}{\partial W^{(1)}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a^{(1)}} \cdot \frac{\partial a^{(1)}}{\partial z^{(1)}} \cdot \frac{\partial z^{(1)}}{\partial W^{(1)}}$$

This process propagates errors backward through the network.

6 Gradient Descent Update

Once gradients are computed, parameters are updated using gradient descent:

$$W \leftarrow W - \eta \frac{\partial L}{\partial W}$$

$$b \leftarrow b - \eta \frac{\partial L}{\partial b}$$

where η is the learning rate.

7 Intuition Behind Backpropagation

Backpropagation answers the question:

How should each weight change to reduce prediction error?

It assigns “credit” or “blame” to each parameter based on its contribution to the final error.

8 Why Neural Networks are Powerful

Neural networks can approximate complex nonlinear functions.

They are particularly useful in finance because:

- Financial data is nonlinear
- Relationships between variables are complex
- Patterns evolve over time

9 Conclusion

Neural networks provide a flexible framework for modeling complex relationships in data.

Backpropagation enables efficient training by computing gradients using the chain rule.

These concepts form the foundation for more advanced architectures such as recurrent neural networks and transformers.